

西南财经大学本科期末考试试题册（A）

（2024—2025学年第1学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（Java SE）(英) 命题教师：王磊

适用对象（年级专业）：2024级特拉华信息管理专业

使用试题的任课教师姓名：王磊

试题说明：

- 1、考试类型：闭卷[] 开卷[☒]
- 2、本套试题共 二 道大题，共 6 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器[] 课本[☒] 等
(请在下划线上填上具体数字或内容，所选[]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
1. 出示学生证（或身份证）和准考证于桌面左上角，以备查验。
 2. 拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 3. 答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。
 4. 所有答案均需填写在答题纸上，答在试题册上无效。
 5. 考生不得携带任何通讯工具进入考场。
 6. 考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。
 7. 严格遵守考场纪律。

Part I: Single Choice Questions:

Mark your answers on below sheet (total 30 pts, 3 pts each)

Answer sheet:

1	2	3	4	5	6	7	8	9	10

1. Which component is used to compile, debug and execute the java programs?_____

- A. JRE
B. JIT
C. JDK
D. JVM

2. Which one cannot be used to declare an array in Java?_____

- A. char
B. String
C. Scanner
D. super

3. Which of the following expressions is equivalent to the boolean expression_____

!(X > 10 && Y != Z)

- A. X > 10 || Y != Z
B. X <= 10 && Y == Z
C. !(X < 10) || (Y != Z)
D. X >= 10 || Y == Z

4. What is the output of the following Java program fragment?_____

```
public static void main(String[] args)
{
    int X = 5;
    int Y = 15;
    update(X, Y);
    System.out.println(X + " " + Y);
}
public static void update (int X, int Y)
```

```
{
    X = X - Y;
    Y = Y - X;
    System.out.println(X + " " + Y);
}
```

- | | |
|-----------|-----------|
| A. 10 25 | B. -10 25 |
| 10 25 | 10 25 |
| C. -10 25 | D. 10 25 |
| 5 15 | 5 10 |

5. What will be the output of the following Java program?_____

```
class Test
{
    public static void main(String args[])
    {
        int arr[] = {1, 2, 3, 4, 5};
        for ( int i = 0; i < arr.length - 2; ++i)
            System.out.println(arr[i] + " ");
    }
}
```

- | | |
|--------------|------------|
| A. 1 2 3 4 5 | B. 1 2 3 4 |
| C. 1 2 | D. 1 2 3 |

6. What will be the output of the following Java program?_____

```
final class A{
    int i=2;
}
class B extends A{
    int j=3;
    public void display(){
        System.out.println(j+" " +i);
    }
}
class Test{
    public static void main(String args[]){
        B obj = new B();
```

```
        obj.display();
    }
}
```

A. 3 2

B. 3 3

C. 2 3

D. Compling error.

7. What will be the output of the following Java program?_____

```
class Square{
    int width;
    int length;
    public Square(int a, int b){
        width=a;
        length=b;
    }
    public int area(){
        return width*length;
    }
}
class Test{
    public static void main(String args[]){
        Square obj= new Square(4,5);
        obj.width=5;
        obj.length=6;
        System.out.println("The area is "+ obj.area());
    }
}
```

A 20

B. 30

C. Runtime error.

D. 0

8. What will be the output of the following Java program?_____

```
class Person{
    public void talk(){
        System.out.print("I am a Person ");
    }
}
class Student extends Person {
```

```
public void talk(){
    System.out.print("I am a Student ");
}
}

public class Test{
    public static void main(String args[]){
        Person p = new Student();
        p.talk();
    }
}
```

- A. I am a Student B. I am a Person
- C. I am a Student D. Output nothing
- I am a Person

The following class **Pizza** has incomplete codes, please complete the class and answer question 9 and 10.

```
public class Pizza {
    String name;
    double diameter;
    void Pizza(){
        // question (9)
    }
    public boolean isLargePizza( )
    {
        // question (10)
    }
}
```

9. The most suitable and correct statement(s) for the constructor is _____.

- A. `super("Italy", 20);`
- B. `super("Italy");`
`super(20);`
- C. `super("Italy");`
`diameter=25;`
- D. `name="Italy";`
`diameter=25;`

10. When the size of the pizza is greater than 20cm, then it must be considered as large pizza. The islargePizza() method must return boolean value indicating whether the pizza is large or

not. Which of the following statement(s) is(are) most suitable for completing the `isLargePizza()` method? _____

- | | |
|--|--|
| A. <code>return diameter>20?true:false;</code> | B. <code>double diameter;</code>
<code>return diameter>20?true:false;</code> |
| C. <code>double diameter;</code>
<code>int result= diameter>20?true:false;</code>
<code>return result;</code> | D. <code>double diameter;</code>
<code>boolean result= diameter>20?true:false;</code>
<code>return result;</code> |

Part II: Write Programs. (total 70 pts)

- (15 pts) For a list of double numbers as below, store them in an array. Then write a Java program to compute the average of them after removing the smallest number. These double numbers are: 9.2 9.5 9.1 9.9 9.0 9.2 9.3 9.4 8.9 9.8.
- (15 pts) Write a program to read a list of grades from the keyboard (integer numbers in the range 0 to 100). Prompt the user with "Please enter a grade between 0 to 100 or -1 to quit: " each time before reading the next integer. Then convert each grade to letter A, B and C as follows: 85 to 100 = A, 70 to 84 = B and 0 to 69 = C. Finally, display the number of A, B and C on the console. For example, if the inputs are:
38 86 92 65 83 72 90 -1
The output should be:
Number of A = 3
Number of B = 2
Number of C = 2
- (15 pts) Design a **Printable** interface, which contains an abstract method, **printInfo()**, to display basic information of an object.

Design an abstract **Car** class, which describes general car objects. It contains:

- Member variable, **brand**, denotes the manufacturer of the car object, e.g. Audi.
- Member variable, **price**, denotes the guide price of the car given by the manufacturer.
- A **constructor**.
- An abstract method, **getSellingPrice()**, returns the actual selling price after discount.

Then, you should design a **UsedCar** class, which extends the **Car** class and implements the **Printable** interface. It contains more members as below:

- Member variable, **age**, denotes the years that the car has been used.
- A constructor, to initialize the member variables of the **UsedCar** object.
- Overridden **getSellingPrice()** method, suppose that the actual selling price of the car reduce 10% annually after being manufactured.
- Overridden **printInfo()** method, to display the brand, selling price, age of the used car.

Finally, you should create a **UsedCar** object in the **main** method, and display its basic information, including its selling price.

4. (25 pts) Design a simple Calculator program to compute the total amount of product sold, as shown in below figure. If the user enters the weight and price of the product, and presses down the “Calculate” button, the total amount will display on the 3rd text field.

Requirements:

- It should have suitable Java FX GUI components and layout.
- It should contain at least one event handling process.

